Semester Project 1

Problem Formulation

Introduction:

Remote-controlled (RC) toy vehicles are a popular option for enthusiasts and amusement, appealing to a broad spectrum of consumers. It is a constant struggle to strike a balance between price, long-lasting performance, and durability. Customers are budget sensitive, particularly in competitive marketplaces, but they also frequently want a product that can endure demanding use and function dependably over time.

Problem Statement:

The key problem is to create an RC toy car that maximizes strength and operating longevity while keeping production costs low. This includes selecting strong materials, optimizing mechanical and electrical components, and implementing efficient production procedures while keeping the finished product affordable to budget-conscious clientele.

Key Criteria:

1. **Durability** – The design of the frame for the car should be able to withstand crashes, bumps and scratches.
2. **Longevity** – Improve the lifespan of the car's battery, motor, and other key components to ensure extended usability.
3. **Cost-Effectiveness** – Identify and implement cost-saving measures in material selection, design, and production without compromising quality.
4. **Profitability** – Maintain a competitive pricing point to attract the target market while attaining acceptable profit margins.
5. **Customer Satisfaction** – Deliver a satisfactory product to the customer and improve reliability.

Additional Criteria:

1. **Safety** – Secure environment for electrical components to prevent overheating and other malfunctions that could harm the client and the product.
2. **Functionality** – Ensuring the programming code executes correctly and delivers consistent performance across a range of user-inputted values.

Delimitation:

Materials such as batteries, motors, wheels, gears, and other electronic components were provided to us, therefore there was no choice for us to decide on them. Furthermore, the libraries and files for the programming were provided to us in a folder that we had to use to implement into the Arduino Nano. Additional components to give the car more functionalities were as a result disregarded as well as libraries and files in visual Studio.

Success Rate:

1. **Durability**
   - Material Strength – Constructing a durable and strong outer frame for the car.
   - Motor Longevity – Designing a safe environment for the motor driver and the Arduino.

2. **Cost Efficiency**
    - Material Costs – Cutting the material cost of the frame and the car to a minimum.
    - Design simplification – Reduce production complexity and make assembly and repair simpler by using modular designs.
3. **Performance**
    - Speed and Maneuverability – Assure responsive and seamless controls at a desired speed.
    - Ease of Use – Create user-friendly controls that are appropriate for the intended age range.
4. **Safety**
    - Age-Appropriate Design – Avoid small parts for younger users and make sure all components are safe for the target age range.
    - Temperature Management – Allowing air to enter and exit the frame to prevent internal heating.

Risk Assessment

A comprehensive risk assessment was conducted to ensure the safe design, assembly, and operation of the car. Key risks identified include programming errors and safety procedures. Programming errors could result in unexpected or erratic behavior of the car, potentially compromising its functionality and performance. Additionally, strict compliance with safety procedures is essential when working with heavy machinery, as improper handling or lack of precautions could lead to serious accidents or equipment damage. A full risk assessment Matrix can be found in *Appendix A.* with description and full assessment criteria *Table 1*.

The time plan for the project is shown as Gantt Chart in *Appendix 2. Chart 1.1*. Generally, we had to make sure that the design was flawless, and the programming was comprehensive with all the components, therefore, we set a lot of time testing, designing, and 3D printing.

Design

The design of the car was kept simple, functional, and with rapid prototyping capability in mind, to fit the semester project requirements. Material selection, design decisions, and iterative testing were done at this stage to ensure that the car performed as it should.

<u>Material and Manufacturing Choices:</u>

We decided not to include more complexity and not complicate the assembly process with an aluminum or even stronger material baseplate; instead, we used one that is 3D printed out of plastic. The reasons are given below:

- **Lightweight Nature:** The car's lightweight design did not demand the durability provided by aluminum, as the car was not intended for strenuous maneuvers or heavy loads.
- **Prototyping Efficiency:** The 3D printer allowed us to make fast developments of the designs and their iterations; hence, this gave a fast turn of adjustments to accommodate the components like the Arduino holder, axle mounting, and drivetrain connections.
- **Integration:** The 3D-printed baseplate integrated key features, including:
    - Axle mountings
    - Arduino and motor driver holders
    - Space for the drivetrain and optocoupler system

Regardless, these advantages came with the drawback of it being brittle in places where the plastic needed to bend to snap on a component. A small example of this is evident in the Arduino holder, where it uses a 'snap-fit' connection like phone cases which must deform while being installed.

Drivetrain and Component Layout:

The drivetrain was designed to utilize the provided metal rods and gears. Key features include:

- **Octocoupler Integration:** A disc with incremental holes was Laser cut and places on the wheel axle. Laser cutting proved to be better than 3d printing for this application because it provides higher accuracy which is needed to for a snug fit on a small scale. The optocoupler fits in-between the brackets and reads these holes, enabling precise speed measurement through laser obstruction timing.
- **Tolerance Testing:** A test piece with varying size holes as shown on *Figure 2.1* was printed to find the right fit of components, ensuring the right level of snugness.

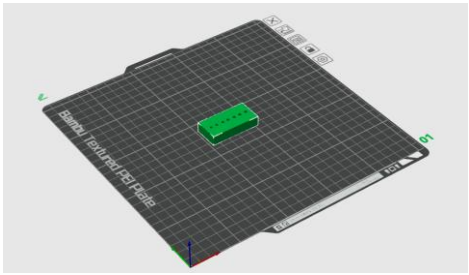


Figure 2.1: Hole Tester for 3D Printer

3D-Printing vs. Aluminum:

Table 2: Material choice evaluation based on our choice of criteria, higher score is better (1-5)

Criteria	Durability	Simplicity	Prototyping Speed	Aesthetic Appeal
3D-Printed Design	2	5	5	4
Aluminum Design	5	3	1	3

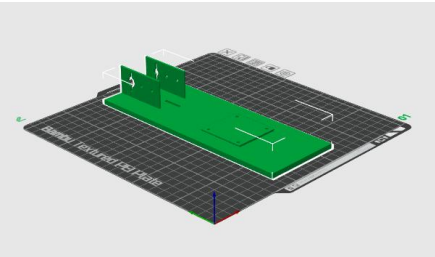


Figure 2.2: Base Design

The 3D-printed design received the highest marks in the simplicity and speed of prototyping. Therefore, ideal for the goals of this project. Durability was less important because the car only needed to operate for the project and did not have to last beyond it.

Aesthetic Considerations:

To improve the looks, a shell modeled of "McQueen" was placed over the car, suspended by a strut in the center. This allowed us to conceal the internal components making the internal design less important visually, meaning we could focus on functionality only. The display was then positioned in McQueen's window, matching with the aesthetic theme.

Challenges and Adjustments:

The primary issues encountered during the design process included:

- **Brittle Plastic:** The 3D-printed plastic occasionally snapped, usually on the snap-fit parts like the Arduino holder.
- **Iterative Prototyping:** Multiple design iterations were needed before achieving the right fit and tolerances.

Overall, despite the challenges, the design phase was quite successful because the car proved functional and met all the requirements of the project. Going for simplicity and rapid prototyping was crucial for this progress of the project as illustrated on figure 1.2 and 1.3.

Testing

After the final assembly shown in *figure 3.1*, we had to test the whole car. First of all, we had to test mechanical parts, so we could be sure that our design is correct, and our car can carry everything we need it to carry.

The first problem was with the motor holder, because it was too loose and because of this issue also the gear connected to the motor did not work properly. We easily fixed this problem just by little editing and printing a new part of this holder. A similar problem was with the wheel holder which should hold the wheel on shaft. But as in the previous case we had to redesign those holders and print them again. In this case it was enough just to make it smaller by 0.1mm. After this edit, the wheels are fixed, and they are properly mounted in the place they should be.

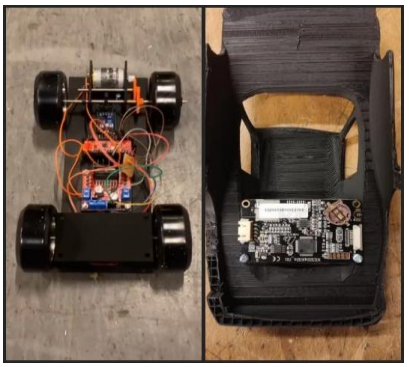


Figure 3.1: Initial Assembly setup

In the next step we had to solve the problem with the holder for the whole case. The problem here was that it did not hold the case straight so on some places it was touching the wheel, so there was problem with smooth movement of the wheel. Here we again needed only one more attempt and after a little editing, every mechanical part was in its proper place.

Unfortunately, it was not that easy with the programming and hardware part. Here we had several problems. We had to solve a problem with the battery indicator, because the program was showing only 0%. We noticed that there was a problem with the connection. Because the program was written differently, than we expected. Because it is much easier to reconnect one wire, than rewrite the program we just reconnected the wire and then we could read the real values.

We tested the rest of the code like PWM regulation and optocoupler counting. Unfortunately, when we put all of the parts together in one complex code as function, we noticed one big problem with our display as shown in *figure 3.2*. Unfortunately, the display was not communicating with our Arduino, so we are not able to write and read any information we need to control our car.



Figure 3.2: Unresponsive Nexion Screen

Final Assessment:

We checked all codes connected with communication with our display and Arduino. We tried to change the setting of I2C bus, we checked connection many times, but we always got the same result. We conclude that the problem has to be somewhere in a deeper communication setting which we did not find.



Figure 3.3: Completed Car 360º view

Conclusion

In summary looking back, the project taught valuable lessons in balancing design simplicity with the need for functionality and durability. An informative table for the final assessment for the project was made in *Appendix 3. Table 3*. While the use of 3D printing enabled rapid development, it became clear that there are trade-offs between material choice, strength, and longevity.

The programming issues, particularly with the communication between the Arduino and the display, highlighted the importance of meticulous debugging and a deeper understanding of component interaction, which could be refined in future projects. In terms of functionality, although the car functioned well mechanically, the programming hiccups did impact the result. If given more time and resources, it would be valuable to focus on refining the software aspect, ensuring seamless integration between hardware and user interface.

In conclusion, while the project was not without its challenges, it provided experience in product design, problem-solving, and working with hardware and software components.

<u>Further investigation:</u>

For further research of the project, we suppose conducting a more in-depth investigation into the programming aspect, as this appeared to be the primary issue causing the car's malfunction and lack of responsiveness. It is crucial to identify and address any errors or inefficiencies in the coding that may have hindered the car's performance. On the other hand, the design and mechanical components of the car were executed with a high degree of precision and appear to function as intended. This focus on programming will ensure a more cohesive integration of all systems in future iterations.

A Appendix 1: Risk Assessment

Table 1: Assessment Criteria

		Likelihood				
		Rare: 1	**Unlikely: 2**	**Possible: 3**	**Likely: 4**	**Almost Certain: 5**
Consequences	**Catastrophe: 5**	4. safety issue	5. Report			
	Major: 4			3. Electrical Issue	2. Programming Errors	
	Moderate: 3		1. Mechanical Failure			
	Minor: 2					
	Negligible: 1					

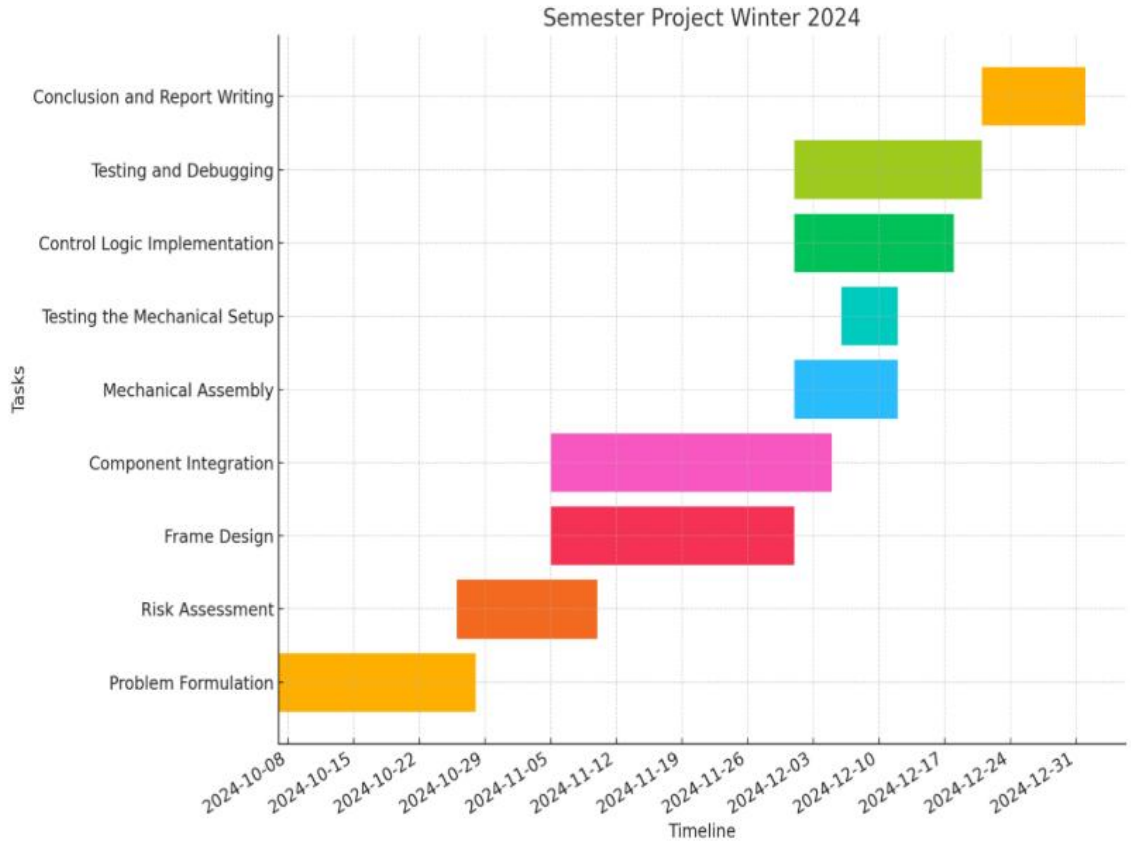
Risk description:

1. Mechanical Failure – Improper fitting of components of the 3D-printed parts.
   Countermeasure: Testing hole plates and precise measuring tools.
2. Programming Errors – The code may cause unexpected car behavior or become unresponsive, increasing risks and prolonging issue identification and debugging.
   Countermeasure: Thoroughly test code and debug iteratively.
3. Electrical Issue - Potential short circuits, overheating of components, and improper connections that could lead to system failures.
   Countermeasure: Ensure proper insulation, secure connections, and monitor components placements for overheating.
4. Safety issue – Handling heavy machinery.
   Countermeasure: Safety protocols for tools like laser cutters and soldering equipment minimize hazards.
5. Report – Report could not possibly be finished on time.
   Countermeasure: Split up work and document throughout the process.

The Risk Assessment table highlights the critical importance of careful programming to prevent faults or errors that could alter default settings, potentially complicating the restoration process. Additionally, maintaining safety in the workspace is extremely important, given the use of heavy machinery. The other risks are significant, but not as important as the other two mentioned.

B Appendix 2: Gantt Chart

Chart 1.1: Gantt chart for time planning



C Appendix 3: Checklist

Table 3: Final Criteria Assessment

Requirement	Met?	Comments
Durability	Partially	The 3D-printed design was lightweight and functional but had brittleness in some components, especially snap-fit parts.
Cost-Effectiveness	Met	The project successfully minimized production costs by using 3D printing and simplifying the design.
Performance	Partially	While mechanical components functioned well, programming issues with the display limited full functionality.
Design Simplicity	Met	The design was kept simple, modular, and easy to assemble, aiding both development and future repairs.
Risk Assessment	Met	A full risk assessment was conducted, addressing potential hazards during assembly and operation.
Safety	Met	Safety measures, like secure battery placement, were successfully implemented.
Display Communication	Not met	The display could not properly communicate with the Arduino, preventing full functionality of the user interface.